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1.) A 200 MHz left-hand circularly polarized plane wave with an electric field modulus of 5 V/m is normally incident in air on a dielectric medium with $\epsilon_r = 4$ and occupies the region defined by $z \geq 0$. (20 points total)

5/5

(a) Write an expression for the electric field phasor of the incident wave given that the field is a positive maximum at $z = 0$ and $t = 0$. (5 points)

$$\tilde{E}^i = a_0 \hat{e} e^{-j k_1 z}$$

$$\tilde{E}^i = 5 (\hat{x} + j\hat{y}) e^{-j \frac{4\pi}{3} z} \text{ V/m}$$

$$f = 200 \times 10^6 \text{ Hz}$$

$$\omega = 2\pi f = 400\pi \times 10^6 \text{ rad/s}$$

$$u = z$$

$$\hat{e} = \text{LHC} = \hat{x} + j\hat{y}$$

$$k_1 = \frac{\omega \sqrt{\epsilon_r}}{c} = \frac{4\pi}{3}$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$a_0 = 5 \text{ V/m}$$

$$\mu_r = \epsilon_r (\text{Air}) = 1$$

(b) Calculate the reflection and transmission coefficients. (5 points)

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$$\eta_1 (\text{Air}) = \sqrt{\frac{\mu_0}{\epsilon_0}} \eta_0 = \sqrt{\frac{1}{1}} (120\pi) = 120\pi$$

$$\eta_0 = 120\pi$$

$$\eta_2 (\text{med. 2}) = \sqrt{\frac{\mu_0}{\epsilon_0}} \eta_0 = \sqrt{\frac{1}{4}} (120\pi) = 60\pi$$

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{60\pi - 120\pi}{60\pi + 120\pi} = \boxed{-\frac{1}{3}} \text{ (unitless)}$$

$$\uparrow = 1 + \Gamma = 1 - \frac{1}{3} = \boxed{\frac{2}{3}} \text{ (unitless)}$$

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(c) Write expressions for the electric field phasors of the reflected wave, the transmitted waves, and the total field in the region $z \leq 0$. (5 points)

$$\tilde{E}^i = \Gamma a_0 \hat{e} e^{\pm j k_1 z}$$

$$\tilde{E}^r = -\frac{1}{3}(5)(\hat{x} + j\hat{y}) e^{+j \frac{4\pi}{3} z} \text{ V/m}$$

$$\tilde{E}^r = -\frac{5}{3}(\hat{x} + j\hat{y}) e^{j(4\pi/3)z} \text{ V/m}$$

$$\tilde{E}^t = \tau a_0 \hat{e} e^{\pm j k_2 z}$$

$$\tilde{E}^t = \tau a_0 \hat{e} e^{\pm j k_2 z}$$

$$= \frac{2}{3}(5)(\hat{x} + j\hat{y}) e^{-j \frac{8\pi}{3} z}$$

$$\tilde{E}^t = \frac{10}{3}(\hat{x} + j\hat{y}) e^{-j \frac{8\pi}{3} z} \text{ V/m}$$

$$\tilde{E}_T = \tilde{E}^i + \tilde{E}^r$$

$$= 5(\hat{x} + j\hat{y}) e^{-j \frac{4\pi}{3} z} - \frac{5}{3}(\hat{x} + j\hat{y}) e^{j \frac{4\pi}{3} z}$$

$$\tilde{E}_T = 5(\hat{x} + j\hat{y}) \left[e^{-j \frac{4\pi}{3} z} - \frac{1}{3} e^{j \frac{4\pi}{3} z} \right] \text{ V/m}$$

$$k_2 = \frac{\omega \sqrt{\epsilon_{r2}}}{c} = \frac{8\pi}{3}$$

$$\epsilon_{r2} = 4$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$\omega = 400\pi \times 10^6 \text{ rad/s}$$

(d) Determine the percentages of the incident average power reflected by the boundary and transmitted into the second medium. (5 points)

5/5

$$\% P_r = 100 |\Gamma|^2 = 100 \left| -\frac{1}{3} \right|^2 = 11.1\%$$

$$\% P_t = 100 |\tau|^2 \left(\frac{n_1}{n_2} \right) = \frac{120\pi}{60\pi} = 88.9\%$$

Check work $P = 11.1\% + 88.9\% = 100\%$, energy conserved ✓

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2. Repeat the above problem but replace the dielectric medium with a poor conductor characterized by $\epsilon_r = 2.25$, $\mu_r = 1$, and $\sigma = 10^{-4} \text{ S/m}$ (20 points total)

$$\omega = 2\pi f = 400\pi \times 10^6 \text{ rad/s}$$

Medium 1: Air $\mu_r = \epsilon_r = 1$

(a) Write an expression for the electric field phasor of the incident wave given that the field is a positive maximum at $z = 0$ and $t = 0$. (5 points)

$$\tilde{E}^i = a_0 \hat{e} e^{Fjk, u} \quad 4.17$$

$$\tilde{E}^i = 5(\hat{x} + j\hat{y}) e^{-j\frac{4}{3}\pi z} \text{ V/m}$$

$$k_1 = \frac{\omega \sqrt{\epsilon_r}}{c} = \frac{400\pi \times 10^6 \sqrt{1}}{3 \times 10^8} = \frac{4}{3}\pi \text{ rad/m}$$

$$u = z \quad a_0 = 5 \text{ V/m} \quad \hat{e} = \text{LHC} = \hat{x} + j\hat{y}$$

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(b) Calculate the reflection and transmission coefficients. (5 points)

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

5/5

$$\eta_1 = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 = 120\pi \Omega$$

$$\eta_2 = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 = \sqrt{\frac{1}{2.25}} 120\pi = 80\pi \Omega$$

$$\Gamma = \frac{80\pi - 120\pi}{80\pi + 120\pi} = \boxed{-\frac{1}{5}}$$

$$\tau = 1 + \Gamma = 1 - \left(-\frac{1}{5}\right) = \boxed{\frac{4}{5}}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$$

$$\omega = 400\pi \times 10^6 \text{ rad/s}$$

$$\epsilon_r = 2.25 \quad \mu_r = 1 \quad \text{Medium 2}$$

$$\sigma = 10^{-4} \text{ S/m}$$

$$\eta_0 = 120\pi$$

$$\epsilon_r = 1 \quad \mu_r = 1 \quad \text{Medium 1 (Air)}$$

(c) Write expressions for the electric field phasors of the reflected wave, the transmitted waves, and the total field in the region $z \leq 0$. (5 points)

$$\tilde{E}^i = \Gamma a_0 \hat{e}^{j k_1 z}$$

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$$= -\frac{1}{5} 5 (\hat{x} + j\hat{y}) e^{j \frac{4}{3} \pi z}$$

$$\tilde{E}^r = -(\hat{x} + j\hat{y}) e^{j \frac{4}{3} \pi z} \text{ V/m} \quad \text{reflective wave}$$

$$\tilde{E}^t = \Gamma a_0 \hat{e}^{j k_2 z} \quad \text{but}$$

$$\frac{\epsilon''}{\epsilon'} = \frac{\delta}{\omega \epsilon_r \epsilon_0} = \frac{10^{-4}}{(400 \pi \times 10^6)(2.25)(8.85 \times 10^{-12})} = 3.99 \times 10^{-3}$$

$$\text{since } \frac{\epsilon''}{\epsilon'} \ll 10^{-2}, \text{ lossless, so } e^{j k_2 z} \rightarrow e^{j \alpha_2 z} e^{j \beta_2 z}$$

$$\alpha_2 = \frac{\delta \mu_0}{2 \sqrt{\epsilon_r}} = \frac{(10^{-4})(120 \pi)}{2 \sqrt{2.25}} = 0.004 \pi$$

$$\beta_2 = \frac{\omega \sqrt{\epsilon_r}}{c} = \frac{400 \pi \times 10^6 \sqrt{2.25}}{3 \times 10^8} = 2 \pi$$

$$\tilde{E}^t = \frac{4}{5} (5) (\hat{x} + j\hat{y}) e^{-j 0.004 \pi z} e^{-j 2 \pi z}$$

$$\tilde{E}^t = 4 (\hat{x} + j\hat{y}) e^{-j 0.004 \pi z} e^{-j 2 \pi z} \text{ V/m} \quad \text{transmitted wave}$$

(d) Determine the percentages of the incident average power reflected by the boundary and transmitted into the second medium. (5 points)

5/5

$$\% P_r = 100 |\Gamma|^2 = 100 \left(-\frac{1}{5}\right)^2 = 4\%$$

$$\% P_t = 100 |\Gamma|^2 \frac{\mu_1}{\mu_2} = 100 \left(\frac{4}{5}\right)^2 \frac{120 \pi}{80 \pi} = 96\%$$

$$\mu_0 = 120 \pi$$

$$\gamma = \frac{4}{5}$$

$$a_0 = 5 \text{ V/m}$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$k_1 = \frac{4}{3} \pi \text{ rad/m}$$

$$\omega = 400 \pi \times 10^6 \text{ rad/s}$$

$$u = z$$

$$\epsilon_r = 2.25$$

$$\Gamma = -\frac{1}{5}$$

$$\hat{e} = \hat{x} + j\hat{y}$$

$$\delta = 10^{-4} \text{ S/m}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$$

$$k_2 = \frac{\omega \sqrt{\epsilon_r}}{c} = \frac{400 \pi \times 10^6 \sqrt{2.25}}{3 \times 10^8} = \frac{3}{4} \pi$$

$$\tilde{E}_{\text{total}} = \tilde{E}^i + \tilde{E}^r$$

$$= 5(\hat{x} + j\hat{y}) e^{-j \frac{4}{3} \pi z} - (\hat{x} + j\hat{y}) e^{j \frac{4}{3} \pi z}$$

$$\tilde{E}_{\text{total}} = (\hat{x} + j\hat{y}) [5 e^{-j \frac{4}{3} \pi z} - e^{j \frac{4}{3} \pi z}] \text{ V/m}$$

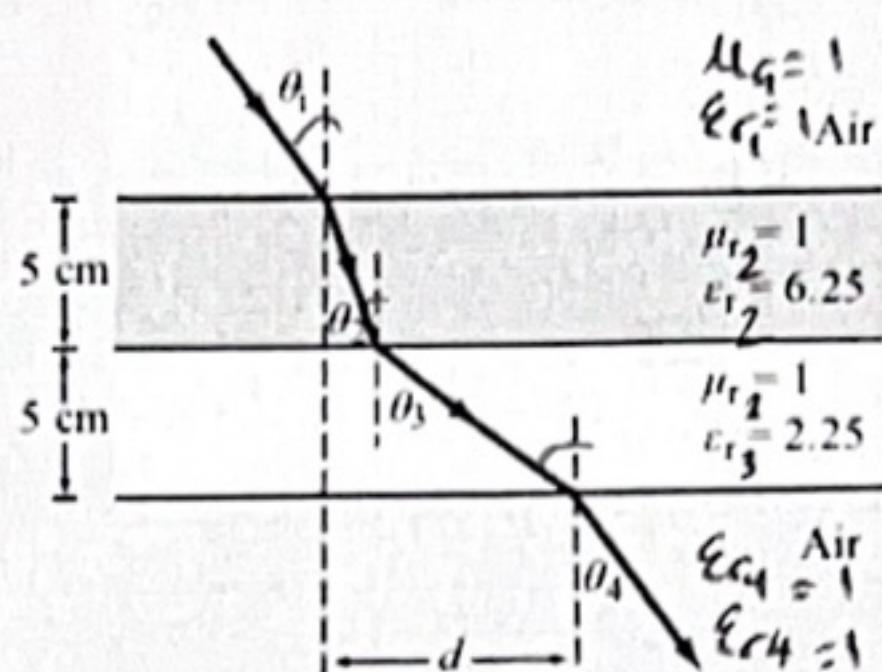
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3. A parallel-polarized plane wave is incident from air at an angle $\theta_i = 30^\circ$ onto a pair of dielectric layers, as shown above. (10 points total)

$$\theta_i = 30^\circ$$

$$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$$

$$\theta_{i+1} = \sin^{-1} \left(\sin(\theta_i) \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}} \right)$$



(a) Determine the angles of transmission θ_2 , θ_3 , and θ_4 . (5 points)

$$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}}$$

$$\sin \theta_2 = \sin(30^\circ) \sqrt{\frac{1}{6.25}}$$

$$\theta_2 = 11.54^\circ$$

$$\theta_4 = \sin^{-1} \left[\sin(\theta_3) \sqrt{\frac{\epsilon_{r3}}{\epsilon_{r4}}} \right]$$

$$= \sin^{-1} \left[\sin(19.47^\circ) \sqrt{\frac{2.25}{1}} \right]$$

$$\theta_4 = 30.00^\circ$$

$$\theta_3 = \sin^{-1} \left(\sin(\theta_2) \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r3}}} \right)$$

$$= \sin^{-1} \left[\sin(11.54^\circ) \sqrt{\frac{6.25}{2.25}} \right]$$

$$\theta_3 = 19.47^\circ$$

(b) Determine the lateral distance d . (5 points)

$$t_2 = t_3 = 0.05 \text{ m}$$

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

$$= t_2 \tan(\theta_2) + t_3 \tan(\theta_3)$$

$$= 0.05 \tan(11.54^\circ) + 0.05 \tan(19.47^\circ)$$

$$d = 0.0279 \text{ m}$$

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4. A TE wave propagating in a dielectric filled waveguide of unknown permittivity has dimensions $a = 5 \text{ cm}$ and $b = 3 \text{ cm}$. If the x component of its electric field is given by

$$E_x = -36 \cos\left(\underbrace{40\pi x}_{\frac{m\pi x}{a}}\right) \sin\left(\underbrace{100\pi y}_{\frac{n\pi y}{b}}\right) \sin\left(\underbrace{(2.4\pi \times 10^{10}) t}_{\omega} - \underbrace{52.9\pi z}_{\beta}\right) \text{ (V/m)},$$

determine: (20 points total)

(a) the mode number, (5 points)

5/5 TE_{mn} (Given from question "TE Wave")

$$\frac{m\pi x}{a} = 40\pi x$$

$$\frac{m}{(0.05)} = 40$$

$$m = 2$$

$$\frac{n\pi y}{b} = 100\pi y$$

$$\frac{n}{0.03} = 100$$

$$n = 3$$

$$TE_{mn} = \boxed{TE_{23}}$$

(b) ϵ_r of the material in the guide, (5 points)

Given: $\beta = 52.9\pi$, $\omega = 2.4\pi \times 10^{10} \text{ rad/s}$, $c = 3 \times 10^8 \text{ m/s}$

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$$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$$

$$= \frac{(3 \times 10^8)^2}{(2.4\pi \times 10^{10})^2} \left[(52.9\pi)^2 + (40\pi)^2 + (100\pi)^2 \right]$$

$$\boxed{\epsilon_r \geq 2.25}$$

(c) the cutoff frequency, (5 points)

5/5 $f_{mn} = \frac{\mu_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ Hz}$

$$f_{23} = \frac{200 \times 10^6}{2} \sqrt{40^2 + 100^2} \text{ Hz}$$

$$= 10.77 \text{ GHz}$$

$$\frac{m}{a} = 40$$

$$\frac{n}{b} = 100$$

$$\mu_{p0} = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{\sqrt{2.25}} = 200 \times 10^6 \text{ m/s}$$

(d) and the expression for $\tilde{H}_y$. (5 points)

5/5 $\tilde{H}_y = \frac{\tilde{E}_x}{Z_{TE}}$

$$Z_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}}$$

$$= 80 \pi$$

$$\sqrt{1 - (10.77 \times 10^9 / 12 \times 10^9)^2}$$

$$= 569.9 \Omega$$

$$\tilde{H}_y = \frac{-36}{569.9} \cos(\dots)$$

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{120 \pi}{\sqrt{2.25}} = 80 \pi$$

$$\omega = 2\pi f$$

$$f = \frac{\omega}{2\pi} = \frac{2.4 \times 10^{10}}{2\pi} = 12 \times 10^9 \text{ Hz}$$

$$\tilde{H}_y = -0.06317 \cos(40\pi x) \sin(100\pi y) \sin(2.4\pi \times 10^{10} t - 52.9\pi z) \text{ A/m}$$

5. A narrow rectangular pulse superimposed on a carrier with a frequency of 9.5 GHz was used to excite all possible modes in a hollow guide with $a = 3$ cm and $b = 2$ cm. If the guide is 100 m in length, how long will it take each of the excited modes to arrive at the receiving end? (8 points)

$$f_c = f_c = 9.5 \times 10^9 \text{ Hz} \quad a = 0.03 \text{ m} \quad b = 0.02 \text{ m}$$

$$f_{mn} = \frac{\mu_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} = \frac{3 \times 10^8}{2} \sqrt{\quad}$$

$$\mu_{p0} = \frac{c}{\sqrt{\epsilon_r}} = 3 \times 10^8 \text{ m/s}$$

Mode	$f_c = f_{mn}$	$\mu_g = \mu_{p0} \sqrt{1 - (f_{mn}/f_c)^2}$	$t = L/\mu_g$
TE ₁₀	$f_{10} = \frac{\mu_{p0}}{2a} = \frac{3 \times 10^8}{2(0.03)}$ $= 5 \text{ GHz} < 9.56$ Excited	$= 3 \times 10^8 \sqrt{1 - \left(\frac{5.6}{9.56}\right)^2}$ $= 255.1 \times 10^6 \text{ m/s}$	$\frac{100}{255.1 \times 10^6}$ $= 0.392 \mu\text{s}$
TE ₀₁	$f_{01} = \frac{\mu_{p0}}{2b}$ $= \frac{3 \times 10^8}{2(0.02)}$ $= 7.5 \text{ GHz} < 9.56$ Excited	$= 3 \times 10^8 \sqrt{1 - \left(\frac{7.56}{9.56}\right)^2}$ $= 184.1 \times 10^6 \text{ m/s}$	$\frac{100}{184.1 \times 10^6}$ $= 543.1 \text{ ns}$
TE ₂₀	$f_{20} = \frac{\mu_{p0}}{a}$ $= \frac{3 \times 10^8}{0.03}$ $= 10 \text{ GHz} > 9.56$ Not excited	N/A Not Excited	NA
TE ₁₁ TM ₁₁	$f_{11} = \frac{\mu_{p0}}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$ $= \frac{3 \times 10^8}{2} \sqrt{\frac{1}{0.03^2} + \frac{1}{0.02^2}}$ $= 9.6 \text{ GHz} < 9.56$ Excited	$\mu_g = \mu_{p0} \sqrt{1 - \left(\frac{9.6}{9.56}\right)^2}$ $= 95.10 \times 10^6 \text{ m/s}$	$\frac{100}{95.10 \times 10^6}$ $= 1.05 \mu\text{s}$